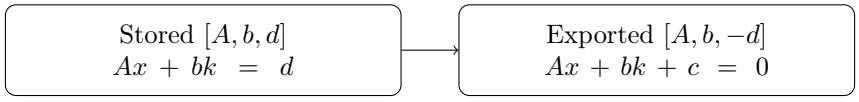
Collector row signs and target information

Offline software result; independent source-fidelity review pending.

The native collector stores $[A, b, d]$ with $d = -a$ and returns d/b after nuisance elimination. The offline information diagnostic declares $Ax + bk + c = 0$. Its corresponding row is therefore $[A, b, -d]$. The adapter changes only that last coefficient.



A sign error can pass every rank check

For the stored row $[0, 1, 5]$ over $\mathbf{F}_7$, the converted homogeneous row is $[0, 1, 2]$ and its target is $k = 5$. Reading the stored row unchanged as homogeneous gives $k = 2$. Both rank triples are $(0, 1, 1)$ and both systems are consistent with a unique target.

More generally, multiplying the entire constant column by -1 preserves rank. The nuisance and nuisance-plus-target matrices are identical; the full matrices differ by an invertible diagonal column transformation. Negation of every unknown gives a bijection between the two homogeneous solution sets. Thus rank, consistency and unique-target booleans alone do not check this sign convention.

Fixed software controls

All 125 one-row triples over $\mathbf{F}_5$ and all 343 over $\mathbf{F}_7$ were enumerated. Complete (x, k) solution sets were checked against a separately written right-hand-side predicate. Six additional fixed systems cover empty, nuisance-only, nonzero target-only, two-row target determination, inconsistent constant-only and target-zero cases. Eight malformed input cases were rejected.

A fresh isolated CLI linked to the existing Rust information diagnostic checked both encodings for each of the 474 systems: 948 outputs. All three ranks, consistency and target-determination outputs matched the explicit solution-set checks. Source inventories were unchanged. These are same-producer cross-implementation checks, not an independent review or a timing comparison. Raw per-case records and the native input/output stream accompany this report.

Source interpretation and boundaries

The source constructs $D = \tau(aG + bH)$ and verifies $D + \sum P_i = 0$ before insertion. To interpret nuisance columns as scalar values, this packet explicitly assumes an additive transfer and a valid homomorphism to $\mathbf{F}_r$ sending $\tau(G)$ to 1 and $\tau(H)$ to k. Then the verified relation gives $Ax + bk + a = 0$. Under that hypothesis, zero-ordinate 2-torsion terms map to zero for odd r. The existence and normalization of this homomorphism, arbitrary point labels, and real retained relation streams are not certified here.

The adapter validates canonical integer residues modulo an integer at least2; admission is not a primality certificate. Rank arguments require a field. The enumerated fields here are exactly 5 and 7. No runtime collector code, scientific workload, cohort result or cost ratio changed.

Custody and next integration step

Sources: `native_solver.rs` lines 471–515, 1058–1119, 1311–1359; existing `rank-information/src/lib.rs`; Sage's primary matrix reference, `https://doc.sagemath.org/html/en/reference/matrices/sage/matrix/matrix2.html`, for right-hand-side equation conventions (Sage itself was not executed).

`source-slices.json` binds exact source bytes. `status.json`, per-field rows, fixed controls, native receipts and input hashes are retained. The editable diagram is `flow.svg`. Next: independent source review, then bind an actual retained stream and its subgroup projection before prospective collector-policy measurement.